Reproducibility.

How to automate access to published data?

What is Scientific Reproducibility?

There is a surprising amount of uncertainty and confusion of what the word **reproducibility** means.

Bioinformatics is a mixture of many sciences - many biologists making use bioinformatics methods don't understand them sufficiently.

Often scientists are unable to articulate the details at the appropriate levels.

A definition of reproducibility

We need to know what someone did. Compare the two statements:

1. *We have quality controlled the data with the tool called* `trimmomatic` *keeping only data with an average quality of 30.*
2. *We ran* `trimmomatic read1.fq SLIDINGWINDOW:4:30` *for quality control.*

Think about the relative merits of each statement.

If you had to redo the analysis which statement would you prefer?

What should be reproducible?

The biological discovery.

It is not about getting the exact same files - we need to obtain results that support the same conclusion.

What if a discovery can only be made with one particular approach?

A Curious Case of Reproducibility

There is an ongoing feud in the RNA-Seq world.

Two software tools appear to be **too similar**. They reproduce each other too well.

The author of the first tool alleges that the second tool is a **reimplemented copy** of their algorithm.

See the lecture page for links to: **accusation** -> **reply to accusation** -> **rebuttal of reply**.

It is a unfortunate and sad story - yet fascinating.

Ingredients of reproducibility

Ensuring that we start with the same data

Ensuring that we know what has been done.

The above can be surprisingly challenging

How to tell if your analysis is reproducible?

Your analysis is reproducible if, and only if,

YOU YOURSELF

are able to reproduce it

at any time, and with ease.

Reproducibility starts RIGHT NOW

Reproducibility is not something you do later, at the end of the analysis.

You should **keep re-doing** it all the time. Erase and start over. Is it hard? Fix the hard part.

A test: If you had to delete all your data and were allowed to keep just a single description of what you did how hard would it be to restore the results? That is "reproducibility".

How do we reproduce data?

We are interested in accessing the data and results for the study titled

- **Genomic surveillance elucidates Ebola virus origin and transmission during the 2014 outbreak**

Published in 2014 in the Science research journal.

A single cryptic line at the very end states:

> *Sequence data are available at NCBI (NCBI BioGroup: PRJNA257197).*

Accession numbers

A "well" defined identifier that links to an entry in a database.

Same information may have multiple access points: `10141003`, `AF086833`, `AF086833.2`

Different databases will use different standards.

Your eye can learn to recognize accession numbers. `PRNJA` will be BioProjects.

NCBI specific numbers

Unique ID : a unique (often numerical, `10141003`) identifier as it is entered into the database. Their use is now discouraged but some tools still use them.

Accession Numbers (Loci): an accession number: `AF086833` applies to the complete database record and remains stable even if updates/revisions are made to the record. The data for a loci **may change**.

Version number: an accession number `AF086833.2` with a version number attached to it. These are unique and data under it **never changes**.

Typical work strategy

1. Familiarize yourself with the data structure. Sometimes it is best done online to visualize it.
2. Write command line code that reproducibly obtains the data for accession numbers.

Project Data:

Resource Name	Number of Links
SEQUENCE DATA	
Nucleotide (Genomic RNA)	249
SRA Experiments	891
Protein Sequences	2240
PUBLICATIONS	
PubMed	1
PMC	1
OTHER DATASETS	
BioSample	716

Using Entrez

Find the page for `PRJNA257197` at the NCBI web-site. Click around on some of the data links. It states:

- `249` genomic features
- `2240` protein sequences

We can get these from the command line with Entrez Direct.

Remember to activate your environment if you get command not found errors:

```
source activate bioinfo
```

Entrez Direct

A command line interface to NCBI. It is a set of tools:

- `esearch` to search databases
- `efetch` to download sequences
- `elink` to connect across databases
- `einfo` to obtain information on databases
- `xtract` to process XML data

See all databases you can search:

```
einfo -dbs
```

Getting data with Entrez Direct

Once you know an accession number you can fetch it in different formats:

```
efetch -db=nuccore -format=gb -id=AF086833 | head -6
```

Produces:

```
LOCUS       AF086833                18959 bp    cRNA    lin
DEFINITION  Ebola virus - Mayinga, Zaire, 1976, complete g
ACCESSION   AF086833
VERSION     AF086833.2
KEYWORDS    .
SOURCE      Ebola virus - Mayinga, Zaire, 1976 (EBOV-May)
```

Same data in other format

We will cover the details of the format in the future lecture.

```
efetch -db=nuccore -format=fasta -id=AF086833 | head -6
```

Produces

```
>AF086833.2 Ebola virus - Mayinga, Zaire, 1976, complete g
CGGACACACAAAAAGAAAGAAGAATTTTTAGGATCTTTTGTGTGCGAATAACTATGAG
TTTTCCTCTCATTGAAATTTATATCGGAATTTAAATTGAAATTGTTACTGTAATCACA
CAGAGCCACATCACAAAGATAGAGAACAACCTAGGTCTCCGAAGGGAGCAAGGGCATC
TGAAAATCCCTTGTCAACACCTAGGTCTTATCACATCACAAGTTCCACCTCAGACTCT
AACAACCTTAATAGAAACATTATTGTTAAAGGACAGCATTAGTTCACAGTCAAACAAG
```

Searching linked information

Our "magic" project number is `PRJNA257197`

```
esearch -db nuccore -query PRJNA257197
```

The search alone reports the counts:

```
<ENTREZ_DIRECT>
  <Db>nuccore</Db>
  <WebEnv>NCID_1_118283973_130.14.18.34_9001_1504627953_20
  <QueryKey>1</QueryKey>
  <Count>249</Count>
  <Step>1</Step>
</ENTREZ_DIRECT>
```

This search could be asked to return `249` results.

The search is database specific

Different databases show different counts:

```
esearch -db protein -query PRJNA257197
```

The search only reports if it has results or not:

```
 <ENTREZ_DIRECT>
        <Db>protein</Db>
        <WebEnv>NCID_1_305224152_130.14.22.215_9001_150463
        <QueryKey>1</QueryKey>
        <Count>2240</Count>
        <Step>1</Step>
</ENTREZ_DIRECT>
```

The search could be asked to return `2240` results.

Obtaining the data

To get the data, pipe the **esearch** into a **efetch**:

```
esearch -db nuccore -query PRJNA257197 | efetch
                                -format fasta > nucleotide.fa
```

What did we fetch:

```
cat nucleotide.fa | head -4
```

Prints nucleotide sequences:

```
>KR105345.1 Zaire ebolavirus isolate Ebola virus/H.sapiens
ATAATTTTCCTCTCATTGAAATTTATATCGGAATTTAAATTGAAATTGTTACTGTAAT
GTTTCAGAGCCATATCACCAAGATAGAGAACAACCTAGGTCTCCGGAGGGGGCAAGGG
CAGTTGAAAATCCCTTGTCAACATCTAGGCCTTATCACATCACAAGTTCCGCCTTAAA
```

Gets database specific data

To get the data, pipe the search into a fetch:

```
esearch -db protein -query PRJNA257197 | efetch
                                -format fasta > proteins.fa
```

What did we fetch:

```
cat proteins.fa | head -3
```

Prints protein sequences:

```
>AKC37233.1 polymerase [Zaire ebolavirus]
MATQHTQYPDARLSSPIVLDQCDLVTRACGLYSSYSLNPQLRNCKLPKHIYRLKYDVT
LPIDFIVPILLKALSGNGFCPVEPRCQQFLDEIIKYTMQDALFLKYYLKNVGAQEDCV
```

Search different databases

Different databases allow different information to be retrieved. All belonging to the same publication.

```
esearch  -db sra -query PRJNA257197 | efetch
                                    -format runinfo > runinfo.csv
```

Now we have a file with lots of columns:

```
cat runinfo.csv | cut -d , -f 1,2,16 | head -3
```

prints:

```
Run,ReleaseDate,LibraryLayout
SRR1972917,2015-04-14 13:59:24,PAIRED
SRR1972918,2015-04-14 13:58:26,PAIRED
```

How to know what information is there?

It is a bit of trial and error. The documentation can be exceedingly obtuse.

```
esearch  -db sra -query PRJNA257197 | efetch
                                    -format summary > summary.xml
```

For example the `runinfo` is a csv file but the `summary` is in XML.

The `summary.xml` is surprisingly large and will download a lot slower.

Why is the NCBI site so hard to use?

It is not just NCBI. The vast majority of repositories are like that.

The scope and needs of life scientists exceed the ability of these organizations to act on that need. Most of them are insular are not required to communicate well.

Being a bioinformatician requires the skill to recognize and work around the endless of limitations and problems that might occur.

You have to learn to **deal with it**.

Programmatic access to other data sources

Each data source may have its own way of accessing the data Programmatically

- **NCBI** - Entrez Direct
- **ENSEMBLE** - REST interface
- **UCSC** - mysql database queries

In this course we will use Entrez Direct.

Note: you will need to manually install `mysql` if you want to try out access to UCSC

Why do we need programmatic access?

To be explicit of what we did. Compare the two:

> *We have obtained the sequence data for project PRJNA257197.*

You still need to figure out how to do it. Versus:

```
# Get nucleotide data.
esearch -db nuccore -query PRJNA257197 | efetch
                                -format fasta > nucleotide.fa
```

See how easily you can get the same data?

Other example: Ensembl REST interface

Search for *Ensembl REST interface* for examples:

```
curl -s
   'https://rest.ensembl.org/lookup/id/ENSG00000157764?'
    -H 'Content-type:application/json'
```

will print:

```
{
"source": "ensembl_havana",
"object_type": "Gene",
"logic_name": "ensembl_havana_gene",
"version": 12,
"species": "homo_sapiens",
"description": "B-Raf proto-oncogene, serine/threonine kin
...
```

Other example: UCSC mysql server

UCSC can also produce data through a `mysql` server.

```
mysql -h genome-mysql.cse.ucsc.edu -u genome -D hg38 -N -A
-e 'select chrom,strand, txStart,cdsStart from knownGene'
```

will produce:

```
chr1    -       17368   17368
chr1    +       29553   29553
chr1    +       30266   30266
...
```

Allows for very sophisticated joins and queries across the tables. You need to understand the data